

ETHAN PARK

📞 647-639-4504 ✉ ethanpk849@gmail.com [in linkedin](#) [🌐 Portfolio](#)

EDUCATION

University of Waterloo

Sep 2023 - Present

Bachelor of Applied Science, Nanotechnology Engineering

Waterloo, ON

- Recipient of the University of Waterloo President's Scholarship
- GPA: 3.7/4.0

EXPERIENCE

Cell Engineering Intern

Sep 2025 - Apr 2026

National Research Council Canada

Mississauga, ON

- Programmed, calibrated, and maintained a robotic liquid-dispensing platform to reliably handle electrode slurry.
- 3D-printed custom hardware to enable the platforms integration to a self driving lab (SDL) for automated battery material synthesis, increasing throughput by 64x.
- Conducted electrochemical testing (EIS, CV, XRD) for 2000+ lithium-ion cells and accelerated analysis by 36x using Python to automate data visualization, coulombic efficiency calculations, and peak finding.
- Achieved specific capacity values within 90% of literature-reported values by investigating NMC phase compositions, dopant concentrations, and PCAM calcination conditions.

Research Assistant (Flow Battery Design)

Apr 2025 - Aug 2025

Van Der Heijden Lab

Waterloo, ON

- Improved mass transport across electrode interfaces by 50% using SolidWorks CFD to optimize flow field designs.
- Reduced electrolyte leakage and mechanical warping by 95% in 3D printed flow cells through iterative optimization of SLA printing parameters, resin selection, and curing processes.
- Investigated nitrogen-assisted carbonization conditions to convert porous polymer structures to functional electrodes.

Electrical Engineering Intern

Jan 2025 - Apr 2025

Smith + Andersen

Toronto, ON

- Drafted electrical device and power distribution drawings in AutoCAD for 15+ public infrastructure projects.
- Completed electrical load and lighting calculations to support energy-efficient layouts, reducing client expenses by 20%.
- Conducted engineering site audits to monitor project progress and incorporated P.Eng. review feedback to validate CAD deliverables for contractors.

EXTRACURRICULAR

Semiconductor Process Engineering Team

May 2026 - Present

Waterloo Hacker Fab

Waterloo, ON

- Leading development of P-type spin-on dopant synthesis using Boric acid and TEOS.
- Designed a test bench for I-V characterization of fabricated semiconductor devices to support electrical testing.
- Investigated thermal oxidation parameters using a custom tube furnace to optimize silicon oxide growth for micro fabrication processes and improve process repeatability.

Class Representative

Sep 2023 - Jan 2024

Waterloo Engineering Society

Waterloo, ON

- Elected to represent the Nanotechnology cohort, serving as a key liaison between students and society executives.
- Gained strong communication and presentation skills by contributing budgetary solutions during society discussions.

TECHNICAL SKILLS

Characterization Techniques: XRD, SEM, EIS, CV, OCP, UV-Vis, I-V

Battery Processing: NMC Cathode Synthesis, Slurry Compounding, Electrode Fabrication, Cell Assembly

Micro Fabrication: Photolithography, Spin Coating, Thin Film Growth, Thermal Oxidation, Cleanroom ISO 4

Programming: Python (Matplotlib, NumPy, SciPy, Pandas), Matlab, Git

CAD and Simulation: SolidWorks, COMSOL, AutoCAD, Simulink